

Overview of Visual question answering

Subject: Visual question answering

1.

Fact-checking if a fact is verified, by posing a question like: is fact X true or false? Customer service, Technical support, Market research, Natural language generation of reports or research.

2.

Further reading Dragomir R. Radev, John Prager, and Valerie Samn. Ranking suspected answers to natural language questions using predictive annotation Archived 2011-08-26 at the Wayback Machine. In Proceedings of the 6th Conference on Applied Natural Language Processing, Seattle, WA, May 2000. John Prager, Eric Brown, Anni Coden, and Dragomir Radev. Question-answering by predictive annotation Archived 2011-08-23 at the Wayback Machine. In Proceedings, 23rd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Athens, Greece, July 2000. Hutchins, W. John; Harold L. Somers (1992). An Introduction to Machine Translation. London: Academic Press. ISBN 978-0-12-362830-5. L. Fortnow, Steve Homer (2002/2003). A Short History of Computational Complexity. In D. van Dalen, J. Dawson, and A. Kanamori, editors, The History of Mathematical Logic. North-Holland, Amsterdam. Tunstall, Lewis (5 July 2022). Natural Language Processing with Transformers: Building Language Applications with Hugging Face (2nd ed.). O'Reilly UK Ltd. p. Chapter 7. ISBN 978-1098136796.

3.

If the right information appears in many forms, the question answering system needs to perform fewer complex NLP techniques to understand the text. Correct answers can be filtered from false positives because the system can rely on versions of the correct answer appearing more times in the corpus than incorrect ones. Some question answering systems rely heavily on automated reasoning.

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interactivity—clarification of questions or answers answer reuse or caching semantic parsing answer presentation knowledge representation and semantic entailment social media analysis with question answering systems sentiment analysis utilization of thematic roles Image captioning for visual question answering Embodied question answering In 2011, Watson, a question answering computer system developed by IBM, competed in two exhibition matches of Jeopardy! against Brad Rutter and Ken Jennings, winning by a significant margin. Facebook Research made their DrQA system available under an open source license. This system uses Wikipedia as knowledge source. The open source framework Haystack by deepset combines open-domain question answering with generative question answering and supports the domain adaptation of the underlying language models for industry use cases. Large Language Models (LLMs)[36] like GPT-4[37], Gemini[38] are examples of successful QA systems that are enabling more sophisticated understanding and generation of text. When coupled with Multimodal[39] QA Systems, which can process and understand information from various modalities like text, images, and audio, LLMs significantly improve the capabilities of QA systems.

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History Two early question answering systems were BASEBALL and LUNAR. BASEBALL answered questions about Major League Baseball over a period of one year. LUNAR answered questions about the geological analysis of rocks returned by the Apollo Moon missions. Both question answering systems were very effective in their chosen domains. LUNAR was demonstrated at a lunar science convention in 1971 and it was able to answer 90% of the questions in its domain that were posed by people untrained on the system. Further restricted-domain question answering systems were developed in the following years. The common feature of all these systems is that they had a core database or knowledge system that was hand-written by experts of the chosen domain. The language abilities of BASEBALL and LUNAR used techniques similar to ELIZA and DOCTOR, the first chatterbot programs. SHRDLU was a successful question-answering program developed by Terry Winograd in the late 1960s and early 1970s. It simulated the operation of a robot in a toy world (the "blocks world"), and it offered the possibility of asking the robot questions about the state of the world. The strength of this system was the choice of a very specific domain and a very simple world with rules of physics that were easy to encode in a computer program. In the 1970s, knowledge bases were developed that targeted narrower domains of knowledge. The question answering systems developed to interface with these expert systems produced more repeatable and valid responses to questions within an area of knowledge. These expert systems closely resembled modern question answering systems except in their internal architecture. Expert systems rely heavily on expert-constructed and organized knowledge bases, whereas many modern question answering systems rely on statistical processing of a large, unstructured, natural language text corpus. The 1970s and 1980s saw the development of comprehensive theories in computational linguistics, which led to the development of ambitious projects in text comprehension and question answering. One example was the Unix Consultant (UC), developed by Robert Wilensky at U.C. Berkeley in the late 1980s. The system answered questions pertaining to the Unix operating system. It had a comprehensive, hand-crafted knowledge base of its domain, and it aimed at phrasing the answer to accommodate various types of users. Another project was LILOG, a text-understanding system that operated on the domain of tourism information in a German city. The systems developed in the UC and LILOG projects never went past the stage of simple demonstrations, but they helped the development of theories on computational linguistics and reasoning. Specialized natural-language question answering systems have been developed, such as EAGLi for health and life scientists. Question answering systems have been extended in recent years to encompass additional domains of knowledge. For example, systems have been developed to automatically answer temporal and geospatial questions, questions of definition and terminology, biographical questions, multilingual questions, and questions about the content of audio, images, and video. Current question answering research topics include:

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rule-based systems, statistical systems, and hybrid systems. Rule-based systems use a set of rules to determine the correct answer to a question. Statistical systems use statistical methods to find the most likely answer to a question. Hybrid systems use a combination of rule-based and statistical methods.

7.

Answering questions related to an article in order to evaluate reading comprehension is one of the simpler forms of question answering, since a given article is relatively short compared to the domains of other types of question-answering problems. An example of such a question is "What did Albert Einstein win the Nobel Prize for?" after an article about this subject is given to the system.

Closed-book question answering is when a system has memorized some facts during training and can answer questions without explicitly being given a context. This is similar to humans taking closed-book exams. Closed-domain question answering deals with questions under a specific domain (for example, medicine or automotive maintenance) and can exploit domain-specific knowledge frequently formalized in ontologies. Alternatively, "closed-domain" might refer to a situation where only a limited type of questions are accepted, such as questions asking for descriptive rather than procedural information. Question answering systems in the context of machine reading applications have also been constructed in the medical domain, for instance related to Alzheimer's disease. Open-domain question answering deals with questions about nearly anything and can only rely on general ontologies and world knowledge. Systems designed for open-domain question answering usually have much more data available from which to extract the answer. An example of an open-domain question is "What did Albert Einstein win the Nobel Prize for?" while no article about this subject is given to the system. Another way to categorize question-answering systems is by the technical approach used. There are a number of different types of QA systems, including: